

Project Progress

9/14/2026

Model Selection

From what I've seen, the reasonable range to try would be 100M up to 2B parameters because they could be run cheaply. In a preliminary way, cost differences in this range seem tiny, so not really something to worry about. I will start testing these:

1. Pythia-160M — very small baseline.
2. Pythia-410M — middle ground.
3. Pythia-1B — upper end of what could work.
 - Pythia models were originally designed in part for interpretability research. Contrasting these 3 models would allow me to figure out what works and doesn't for this project based on parameter count alone, while the architecture, training data, etc., stays the same.
 - More info on Pythia models:
 - https://huggingface.co/EleutherAI/pythia-160m?utm_source=chatgpt.com
 - Official paper on Pythia release: <https://arxiv.org/pdf/2304.01373>
4. Qwen2.5-0.5B — More modern architecture with RoPE, SwiGLU, RMSNorm and grouped-query attention; it gives something architecturally different to evaluate.
5. Either GPT-2 Small as a deliberately simple old-school reference, or something like SmolLM2-360M as another modern compact model.

model_selection.ipynb

1. Load Candidate models
2. Create common test prompts/settings
3. Baseline generation
4. Runtime + memory measurements
5. Simple neuron intervention
6. Simple attention intervention
7. Comparison table
8. Model-selection conclusion

API contract

We would need to agree on a minimal version of this and define it. It could be changed as we add or modify things.

The front end could send an HTTP POST request with a JSON body like:

```
1 {  
2   #The user's prompt for the model  
3   "prompt": "The capital of France is",  
4   #Generation parameters decided by user  
5   "generation": {  
6     "max_new_tokens": 50,  
7     "do_sample": false,  
8     "temperature": 0.8,  
9     "top_k": 50,  
10    "top_p": 0.95  
11  },  
12  #Inference time intervention parameters decided by user  
13  "intervention": {  
14    "type": "mlp_ablation",  
15    "layer": 0,  
16    "proportion": 0.5  
17  }  
18 }
```

The backend would return a JSON HTTP response that allows displaying the text generated by the modified and unmodified model side-by-side, plus whatever intervention was performed:

```
1 {
2   #Text generated by model without interventions
3   "baseline_text": "...",
4   #Text generated by model with with user's interventions
5   "modified_text": "...",
6   #Information about interventions, could be displayed somewhere.
7   "intervention": {
8     "type": "mlp_ablation",
9     "layer": 0,
10    "proportion": 0.5
11  }
12 }
```